Section 5.4 Expected values & stopping processes

期望与停止过程

Method 1 / 方法 1 [PS04-M01]

An expectation is a weighted total, not necessarily a typical outcome. Use symmetry for exchangeable gaps, tails for integer scores, and complete states for stopping processes. A strategy is locally optimal only when its choice does not change the future state distribution.

期望是加权总量，不一定是典型结果。可交换间隙用对称性，整数分数用尾概率，停止过程用完整状态。只有决策不影响未来状态分布时，逐步贪心才自动最优。

Average order statistics through exchangeable gaps

用可交换间隙求顺序统计量均值

Procedure

1. Write the selected positions increasingly and record all unselected gaps, including both end gaps.

2. Prove a bijection to weak compositions of the unselected total; uniform subsets then give exchangeable gaps.

3. Each gap has the same mean, and their means sum to the fixed total.

4. Express the desired ordered position as its rank plus the preceding gaps.

通用步骤

1. 将所选位置递增排列，记录所有未选间隙，含首尾间隙。

2. 证明它们与未选总量的非负整数组成双射；均匀子集于是给出可交换间隙。

3. 各间隙均值相同，且均值总和等于固定未选总量。

4. 第几小的位置等于其序号加上它前面的所有间隙。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 1 / 方法 1 [PS04-M01] (continued / 续)

> Example. Choose four positions uniformly from 1 through 9. Find the expected second-smallest position.

> 例题：从一至九中等可能选四个位置，求第二小位置的期望。

Worked solution

1. The five unchosen positions form five nonnegative gaps $g_0, \ldots, g_4$ with sum five. Each weak composition occurs once.

2. Exchangeability gives $E(g_i) = 1$ for every gap.

3. The second chosen position is $2 + g_0 + g_1$, so its expectation is 4.

例题解析

1. 五个未选位置形成五个非负间隙 $g_0, \ldots, g_4$，总和五，每种非负组成恰对应一个子集。

2. 由可交换性，每个间隙的期望为一。

3. 第二个所选位置为 $2 + g_0 + g_1$，故期望为四。

The gaps are exchangeable, not independent; equal means need only symmetry.

间隙可交换但并不独立；求相同均值只需对称性。

Practice / 配套练习: AIME-2015-I-Q12 (WB p. 777)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 2 / 方法 2 [PS04-M02]

Sum tail probabilities for an integer score

对整数分数累加尾概率

Procedure

1. For a nonnegative integer score X, use $E(X)=\sum_{j\ge 1}\Pr(X\ge j)$.

2. If a lower bound L is automatic, start with L and sum only tails above it.

3. Translate each tail event into simultaneous restrictions and count it in the original uniform sample space.

4. For nested allowed sets, assign the smallest set first and subtract already-used choices.

通用步骤

1. 非负整数分数满足 $E(X)=\sum_{j\ge 1}\Pr(X\ge j)$。

2. 若自动有下界 L，先记 L，只累加更高的尾概率。

3. 把每个尾事件转为同时满足的限制，并在原等可能样本空间中计数。

4. 允许集合嵌套时，先分配最小集合，再减去已使用的选择。

> Example. Roll two independent fair dice. Find the expected larger result.

> 例题：独立投两枚公平骰子，求较大点数的期望。

Worked solution

1. Let X be the maximum. For $1\le j\le 6$, $\Pr(X\ge j)=1-((j-1)/6)^2$.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 2 / 方法 2 [PS04-M02] (continued / 续)

2. Therefore $E(X) = 6 - (0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2)/36$.

3. The square sum is 55, so the expected maximum is $161/36$.

例题解析

1. 令 X 为最大点数，则 $1 \le j \le 6$ 时，$\Pr(X \ge j) = 1 - ((j-1)/6)^2$。

2. 所以 $E(X) = 6 - (0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2)/36$。

3. 平方和为 55，故最大值期望为 $161/36$。

A minimum exceeding a threshold means every component passes it, which often makes tails easier than the full distribution.

最小值超过门槛等价于所有分量都过关，故尾概率往往比完整分布好数。

The identity is pointwise: $X = \sum_{j \ge 1} \mathbf{1}_{X \ge j}$. For a rook assignment with nested row-allowed sets $S_1 \subseteq \cdots \subseteq S_r$, earlier choices all belong to the next set, so row i has exactly $|S_i| - (i-1)$ available columns. Complete any unrestricted rows by a factorial.

尾和恒等式来自逐点等式 $X = \sum_{j \ge 1} \mathbf{1}_{X \ge j}$。车的行列分配若允许集合满足 $S_1 \subseteq \cdots \subseteq S_r$，之前所选列均属于下一集合，故第 i 行恰有 $|S_i| - (i-1)$ 个可用列；余下无限制行用阶乘排列。

Practice / 配套练习: AIME-2016-II-Q13 (WB p. 779)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03]

Write first-step equations with the correct boundary reward

按边界奖励写首步方程

Procedure

1. Choose states that determine every future transition, including holding and reflecting behavior.

2. A hitting probability satisfies $p_s = \sum_t P_{st} p_t$ with boundary values zero or one.

3. An expected step count satisfies $T_s = 1 + \sum_t P_{st} T_t$ with zero at absorption; keep any self-transition term.

4. Solve the finite system and justify absorption or finite expectation using a uniform positive chance of success in bounded blocks.

通用步骤

1. 选择能决定所有未来转移的状态，包括原地停留及反射行为。

2. 命中概率满足 $p_s = \sum_t P_{st} p_t$，边界值为零或一。

3. 期望步数满足 $T_s = 1 + \sum_t P_{st} T_t$，吸收状态为零；原地转移项不能漏。

4. 解有限方程组，并用有限步块中统一的正成功概率证明吸收或期望有限。

> Example. A walker on $0, 1, 2, 3$ moves right with probability $2/3$ and left with $1/3$, stopping at either endpoint. Starting at one, find the probability of stopping at three and the expected number of moves.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03] (continued / 续)

> 例题：行者在 $0,1,2,3$ 上移动，向右概率 $2/3$、向左 $1/3$，到任一端点就停止。从一出发，求停在三的概率及期望步数。

Worked solution

1. The hitting probabilities obey $p_1 = 2p_2/3$ and $p_2 = 2/3 + p_1/3$. Thus $p_1 = 4/7, p_2 = 6/7$.

2. The expected times obey $T_1 = 1 + 2T_2/3$ and $T_2 = 1 + T_1/3$. Solving gives $T_1 = 15/7, T_2 = 12/7$.

3. From either interior state, two consecutive right moves guarantee absorption with probability at least $4/9$. Thus the expected stopping time is finite, validating these equations.

例题解析

1. 命中概率满足 $p_1 = 2p_2/3, p_2 = 2/3 + p_1/3$，解得 $p_1 = 4/7, p_2 = 6/7$。

2. 期望满足 $T_1 = 1 + 2T_2/3, T_2 = 1 + T_1/3$，解得 $T_1 = 15/7, T_2 = 12/7$。

3. 从任一内点连续两次右行都保证已吸收，概率至少 $4/9$，故停止期望有限，以上方程有效。

Probability equations have boundary values; time equations have a unit cost per step. Do not interchange them.

概率方程靠边界值，时间方程每步另加一，不能混用。

For a height walk with equally likely east, west, north, south, the two horizontal holding moves give $T_{j+1} - 2T_j + T_{j-1} = -4$. At a reflecting boundary where only east, west, north are available, $T_0 = 1 + (2T_0 + T_1)/3$, hence $T_0 - T_1 = 3$. First differences reduce the interior equation to an arithmetic sequence.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03] (continued / 续)

For a strictly increasing walk, a target can be visited at most once. Partitioning paths by the last step gives a visit-probability recurrence without an expected-time unit cost.

若高度行走等可能向东西南北，两种水平停留使 $T_{j+1} - 2T_j + T_{j-1} = -4$。反射边界若只可向东西北，则 $T_0 = 1 + (2T_0 + T_1)/3$，故 $T_0 - T_1 = 3$。对期望作一阶差分，内点方程便化为等差数列。

严格向前的行走至多访问目标一次，因此可按最后一步分类得到访问概率递推；这里不能额外加期望时间的单位代价。

Practice / 配套练习: AIME-2016-I-Q13 (WB p. 778), AIME-2019-II-Q02 (WB p. 774), AIME-2021-I-Q12 (WB p. 781)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 4 / 方法 4 [PS04-M04]

Aggregate symmetric states only after checking transition totals

核对转移总量后才能合并对称状态

Procedure

1. Replace absolute positions by relative gaps or another symmetry invariant.

2. Check that every microscopic state in one proposed class has the same probabilities of moving to every class.

3. Enumerate all simultaneous elementary moves, including collisions and any memory restrictions.

4. Use the resulting class transition table in the first-step equations.

通用步骤

1. 以相对间隙等对称不变量代替绝对位置。

2. 核对同一拟定类别中的每个细状态，转向各类别的概率均相同。

3. 枚举同时发生的全部基本移动，包括碰撞和历史限制。

4. 把所得类别转移表代入首步方程。

> Example. Two walkers start at opposite vertices of a square. Each second both independently choose a neighboring vertex with equal probabilities. They stop on their first simultaneous meeting. Find the expected stopping time.

> 例题：两名行者从正方形对顶点出发，每秒同时、独立等可能选一个相邻顶点，到首次同时相遇时停止。求停止时间期望。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 4 / 方法 4 [PS04-M04] (continued / 续)

Worked solution

1. Before meeting they remain opposite: equal clockwise choices keep them opposite, while opposite choices send them to the same vertex.

2. Of the four equally likely simultaneous choices, two meet and two return to the same opposite-state class.

3. For expected time T, $T = 1 + T/2$, so $T = 2$. The geometric survival probability 2^{-n} also proves finite expectation.

例题解析

1. 未相遇时两人仍在对顶点；同向转动仍对顶，反向选择则在同一顶点相遇。

2. 四种等可能同时选择中，两种相遇，两种返回相同的对顶状态类。

3. 期望 T 满足 $T = 1 + T/2$，故为两秒；存活概率 2^{-n} 也证明期望有限。

Equal-looking states can have different transition counts; symmetry must preserve the process, not only the drawing.

外观相同的状态可能转移数不同；所用对称必须保持过程，而不只是图形。

Practice / 配套练习: AIME-2021-I-Q12 (WB p. 781)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 5 / 方法 5 [PS04-M05]

Use a signed generating value for stopping-time parity

用带符号生成值求停止次数奇偶

Procedure

1. Track the longest suffix that matches a prefix of the target pattern.

2. Let $u_s = E((-1)^R)$ for remaining rolls. Every additional roll contributes a factor -1, and a completed pattern has remaining value one.

3. Write the first-step equations, including fallback to shorter matched prefixes.

4. Recover $\Pr(R \text{ odd}) = (1 - u_s)/2$ after proving almost-sure stopping.

通用步骤

1. 以“当前结尾能匹配目标前缀的最长长度”为状态。

2. 令 $u_s = E((-1)^R)$，每增加一步乘负一，已完成状态的余下次数为零、对应值为一。

3. 写首步方程，并包含退回较短匹配前缀的情况。

4. 证明几乎必然停止后，用 $\Pr(R\text{为奇数}) = (1 - u_s)/2$ 求概率。

> Example. Toss a fair coin until the first consecutive pattern HT. Find the probability the total toss count is odd.

> 例题：投公平硬币，首次连续出现 HT 时停止，求总次数为奇数的概率。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 5 / 方法 5 [PS04-M05] (continued / 续)

Worked solution

1. Use state zero for no trailing H, and state one for a trailing H. Let the corresponding signed expectations be u_0, u_1.

2. First-step equations are $u_0 = -(u_0 + u_1)/2$ and $u_1 = -(u_1 + 1)/2$. Thus $u_1 = -1/3, u_0 = 1/9$.

3. The desired probability is $(1 - 1/9)/2 = 4/9$. Disjoint two-toss blocks have a fixed $1/4$ chance of HT, so stopping is almost sure.

例题解析

1. 用零状态表示末尾不为 H，一状态表示末尾为 H，带符号期望分别为 u_0, u_1。

2. 首步方程为 $u_0 = -(u_0 + u_1)/2, u_1 = -(u_1 + 1)/2$，解得 $u_1 = -1/3, u_0 = 1/9$。

3. 所求概率为 $(1 - 1/9)/2 = 4/9$。互不重叠的两次块各有 $1/4$ 概率出现 HT，故几乎必然停止。

Immediate completion still uses one roll, so its contribution has a minus sign.

立即完成仍消耗一步，因此在方程中的贡献带负号。

Practice / 配套练习: AIME-2018-II-Q13 (WB p. 780)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 6 / 方法 6 [PS04-M06]

Optimize a prediction when choices do not affect future states

决策不影响未来状态时优化即时预测

Procedure

1. Prove that the current guess or action does not alter the next-state distribution.

2. Maximize the conditional immediate expected reward at each state.

3. Add that reward to the transition-weighted future value; keep terminal states exact.

4. Use a two-parameter recurrence when the remaining counts determine the state.

通用步骤

1. 先证明当前猜测或行动不会改变下一状态分布。

2. 在每个状态最大化当前条件期望奖励。

3. 将该奖励加上按转移概率加权的未来价值，并保留准确边界状态。

4. 若剩余数量决定状态，就用双参数递推。

> Example. A shuffled deck has two red cards and one black card. Before each reveal, a correct red guess earns two points and a correct black guess one point. Find the maximum expected total reward.

> 例题：随机洗匀两红一黑牌，每次翻牌前猜色，猜中红色得两分、黑色得一分。求最大期望总得分。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 6 / 方法 6 [PS04-M06] (continued / 续)

Worked solution

1. At state (r, b), choose red when $2r \geq b$. The guess does not affect which card is revealed.

2. Let F be optimal expected reward. Then $F(1,0) = 2, F(0,1) = 1, F(2,0) = 4$, and $F(1,1) = 1 + (1+2)/2 = 5/2$.

3. At $(2,1)$, the immediate expectation is $4/3$, so $F(2,1) = 4/3 + (2/3)(5/2) + (1/3)4 = 13/3$.

例题解析

1. 状态 (r, b) 下，若 $2r \geq b$ 就猜红；猜测并不改变翻出的牌。

2. 令 F 为最优期望，边界 $F(1,0) = 2, F(0,1) = 1, F(2,0) = 4$，且 $F(1,1) = 1 + (1+2)/2 = 5/2$。

3. 在 $(2,1)$ 中，即时奖励期望为 $4/3$，故 $F(2,1) = 4/3 + (2/3)(5/2) + (1/3)4 = 13/3$。

Majority color is optimal for equal correct-guess rewards; unequal rewards change the threshold.

只有猜中两色奖励相同时才直接选多数色；奖励不等会改变阈值。

Practice / 配套练习: AIME-2023-I-Q06 (WB p. 775)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 7 / 方法 7 [PS04-M07]

Count regions by linearity of crossing indicators

用交叉指示量的线性性数区域

Procedure

1. When adding a segment with distinct interior crossings, its new regions equal one plus its number of crossings.

2. Express the total as a fixed segment contribution plus one indicator for each crossing pair.

3. Compute each pair probability by endpoint interleaving under the actual conditioned endpoint distribution.

4. Use linearity; pairwise crossing indicators need not be independent. Treat prescribed multiple crossings separately.

通用步骤

1. 加入线段时，若内部交点互异，新增区域数等于交点数加一。

2. 将总量写为固定线段贡献加上每对交叉的指示量。

3. 根据真实的条件端点分布，用端点交错求各对交叉概率。

4. 使用期望线性性，无须交叉事件独立；预设的多线共点须另行处理。

> Example. A disk has two fixed perpendicular diameters and three independent chords, each formed by two uniform independent perimeter endpoints. Find the expected number of regions.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 7 / 方法 7 [PS04-M07] (continued / 续)

> 例题：圆盘内有两条固定垂直直径，再独立画三条弦，每弦两端独立均匀取自圆周，求期望区域数。

Worked solution

1. There are five segments. Apart from probability-zero coincidences, regions equal $1+5+$ the number of crossing pairs. The fixed diameters contribute one crossing.

2. Each random chord crosses a given diameter with probability $1/2$, since its endpoints must lie in opposite semicircles. Two independent random chords cross with probability $1/3$: for four cyclically ordered endpoints, exactly one of the three pairings interleaves.

3. Thus the expectation is $6+1+3\cdot 2\cdot(1/2)+\binom{3}{2}(1/3)=11$.

例题解析

1. 共有五条线段。除零概率重合外，区域数等于 $1+5+$交叉线段对数；固定直径贡献一个交点。

2. 随机弦与一条直径相交概率为 $1/2$，因为两端须在不同半圆；两随机弦相交概率为 $1/3$，四个圆周顺序端点的三种配对中恰一种交错。

3. 所以期望为 $6+1+3\cdot 2\cdot(1/2)+\binom{3}{2}(1/3)=11$。

Conditioning endpoints to different quadrants changes the crossing law; do not reuse the unrestricted one-third value.

若端点被限制在不同象限，相交概率会改变，不能沿用无条件的三分之一。

Practice / 配套练习: AIME-2025-I-Q13 (WB p. 782)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 8 / 方法 8 [PS04-M08]

Stop at the first third-category event and sum completed prefixes

在第三类首次出现时停止并累加已完成前缀

Procedure

1. Partition trials into two desired categories and a stopping category.

2. Condition on the full pre-stop word, including its order and the final stopping event.

3. Count or weight words that complete both desired quotas; handle overlap of deficient-category events.

4. Sum the resulting geometric or arithmetic-geometric series and check the starting index.

通用步骤

1. 把试验分为两种目标类别及一种停止类别。

2. 按停止前的完整词分类，保留顺序并计入最后的停止事件。

3. 对达到两种配额的词计数或加权；缺额事件可能有交集。

4. 累加等比或等差乘等比级数，并核对起始下标。

> Example. Independent trials produce A, B, C with probabilities $1/2, 1/4, 1/4$. What is the chance both A and B appear before the first C?

> 例题：独立试验出现 A, B, C 的概率为 $1/2, 1/4, 1/4$。求首次出现 C 之前，A, B 都已出现的概率。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 8 / 方法 8 [PS04-M08] (continued / 续)

Worked solution

1. No A before the first C means a word B^kC, with total probability $\sum_{k\ge0}(1/4)^{k+1}=1/3$.

2. No B similarly has probability $\sum_{k\ge0}(1/2)^k(1/4)=1/2$. Their intersection is an immediate C, with probability $1/4$.

3. By inclusion-exclusion, success is $1-1/3-1/2+1/4=5/12$. The final stopping-event weight is included in both series.

例题解析

1. 首次 C 之前无 A，对应词 B^kC，总概率为 $\sum_{k\ge0}(1/4)^{k+1}=1/3$。

2. 无 B 的概率同理为 $\sum_{k\ge0}(1/2)^k(1/4)=1/2$，两者交集为立即出现 C，概率 $1/4$。

3. 容斥得成功概率 $1-1/3-1/2+1/4=5/12$；两级数都计入了末次停止事件的权重。

For two quotas of two, deficient binary counts are disjoint only once the prefix length is at least four.

两类配额各为二时，只有前缀长至少四，两个缺额二元词类才互斥。

For $|x|<1$, $\sum_{k\ge0}x^k=1/(1-x)$ and $\sum_{k\ge0}(k+1)x^k=1/(1-x)^2$. Subtract the finitely many terms before the valid quota length. The second identity follows by multiplying two geometric series, so differentiation is optional.

当 $|x|<1$，有 $\sum_{k\ge0}x^k=1/(1-x)$ 及 $\sum_{k\ge0}(k+1)x^k=1/(1-x)^2$。减去达到配额前的有限项即可；第二式也可由两个等比级数相乘得到，无须使用求导。

Practice / 配套练习: AIME-2026-II-Q07 (WB p. 776)